

CONTROLLED
CONDITIONS

Shared data and features

WIN 5-minute futures | 15,057 bars | Information Gain: 7 features



Shared temporal protocol

60% training | 20% validation | 20% locked test | no shuffling



Shared search control

RS, GA, PSO, DE, GWO | four backbones | 1,500 evaluations per seed



Protocol A: MCC/ F_1 fitness

Validation objective
 $0.60 \text{ MCC} + 0.40 F_1$

Protocol B: weighted-accuracy fitness

Weighted training/validation accuracy

OBJECTIVE
VARIES



Common locked evaluation

Test MCC, F1, and accuracy reported after selection
No test-set feedback during search